Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/20/2021
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/10/2017	**End Date:**	06/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2022 per F. Hernandez/ARC (Ed., 8/19/21) NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/2020) NOTE: Extended to 7/31/2020 per NSSC information (Ed., 5/15/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC17K0546		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	[ED. NOTE 12/11/2017: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants--NNX12AO65G," per PI move from University of Mississippi to University of North Carolina at Greensboro.] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments.		

	These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will be launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying the effects of gravity and light on basic cellular mechanisms in plants, this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	Traveling to nearby extraterrestrial objects having a reduced gravity level (partial gravity) compared to Earth's gravity is becoming a realistic objective for space agencies. The use of plants as part of life support systems will require a better understanding of the interactions among plant growth responses (tropisms) under partial gravity conditions. Here we show results from our latest space experiments on the ISS, in which seeds of Arabidopsis thaliana were germinated, and seedlings grew for six days under different gravity levels, namely micro-g, several intermediate partial-g levels, and 1g, and were subjected to irradiation with blue light for the last 48 hours. RNA was extracted from 20 samples (pools of 10-15 seedlings each) for subsequent RNAseq analysis. Transcriptomic analysis was performed using the HISAT2-Stringtie-DESeq pipeline. Differentially expressed genes were further characterized for global responses using GEDI tool, gene networks and for Gene Ontology (GO) enrichment. Differential gene expression analysis revealed only one differentially expressed gene across all gravity conditions using false discovery rate ($q < 0.05$). However, the same 14 genes appeared differentially expressed when comparing either micro-g, the Moon g-level, or the intermediate low-g level (< 0.1g) with 1g control conditions. Apart from these 14-shared genes, the number of variations was similar in microgravity and the Moon g-level and increased in the intermediate g-level (< 0.1g), but it was then progressively reduced as the difference with the Earth gravity became smaller. The GO groups were differentially affected at each g-level: light and photosynthesis GO under microgravity, genes belonged to general stress, chemical, and hormone responses under low-g and a response more related to cell wall and membrane structure and function under the Moon g-level. Plant transcriptional response under blue light stimulation suggests that recently described root blue-phototropism may be enough to reduce the gravitational stress response caused by the lack of gravitropism on orbit. Competition between tropisms induces an intense perturbation at the micro-g level, which shows an extensive stress response that is progressively attenuated. A conspicuous effect on cell wall/membrane remodeling, detected at the interval from the Moon to Mars gravity, can be potentially related to graviresistance mechanisms. Our other spaceflight studies show the dysregulation of different hormonal routes in our samples and suggest that, in microgravity, the seedlings do not address the external stress and take the option of growing by activating growth and proliferation-promoting auxin and cytokinin routes. On the other hand, in the Mars-g samples, the growth-promoting routes are inhibited and abscisic acid (ABA), ethylene, and salicylic acid routes, known for their crucial role in response to osmotic, drought, and biotic stresses, are activated. These studies show the utility of spaceflight experiments in making discoveries in the fundamental biology of plants, which will be needed to fully utilize plants in bioregenerative life support systems in space missions.
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